

LOGICAL REASONING

Analogy

Analogy is one type of logical reasoning question where you identify the relationship between two things and apply the same relationship to another pair.

Basic Format

$$A : B :: C : ?$$

Read it as:

A is related to B in the same way that C is related to?

Type of Analogy Problem:

1. Number Analogy
2. Alphabet Analogy
3. Word Analogy
4. Mixed Analogy

1. Number Analogy

Example:

$$8 : 72 :: 10 : ?$$

First find the relationship:

$$8 \times 9 = 72$$

So apply the same to 10:

$$10 \times 11 = 110$$

Answer = **110**

How to approach Number Analogy

Check operations in this order:

Step 1 — Addition/Subtraction

Example:

$$7 : 12$$

Relationship:

$$7 + 5 = 12$$

Step 2 — Multiplication/Division

Example:

$$8 : 48$$

$$8 \times 6 = 48$$

Step 3 — Square/Cube

Example:

$$6 : 36$$

$$6^2 = 36$$

or

$$4 : 64$$

$$4^3 = 64$$

Step 4 — Combination

Example:

$$6 : 42$$

Possible relationship:

$$6 \times 7 = 42$$

or

$$6^2 + 6 = 42$$

You must check the **second pair** to determine which relationship is intended.

Common Number Patterns

Learn to check:

- $+ n$
- $- n$
- $\times n$
- $\div n$

- n^2
- n^3
- $n^2 + n$
- $n^2 - n$
- $n(n + 1)$
- $n(n - 1)$
- $2n + 1$
- $2n - 1$
- $n(n + 2)$
- combinations of two operations

2. Alphabet Analogy

Alphabet questions require you to convert letters into their **positions**.

Alphabet positions

A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

Example

C : F :: H : ?

Convert:

$$C = 3$$

$$F = 6$$

Relationship:

$$+3$$

Therefore:

$$H = 8$$

$$8 + 3 = 11$$

$$11 = K$$

Answer = **K**

What to check in Alphabet Analogy

Pattern 1 — Forward movement

B : E :: G : ?

$$B \rightarrow E = +3$$

$$G \rightarrow J = +3$$

Ans: **J**

Pattern 2 — Backward movement

T : Q :: M : ?

$$T \rightarrow Q = -3$$

$$M \rightarrow J = -3$$

Ans: **J**

Pattern 3 — Reverse alphabet

Remember:

$$A \leftrightarrow Z$$

$$B \leftrightarrow Y$$

$$C \leftrightarrow X$$

$$D \leftrightarrow W$$

Example:

C : X :: F : ?

C's opposite = X

F's opposite = U

Ans: U

Pattern 4 — Multiple letters

Example:

ABC : DEF :: GHI : ?

Each letter moves +3.

A → D

B → E

C → F

Therefore:

G → J

H → K

I → L

Ans : **JKL**

Important trick

For alphabet questions, immediately write:

A=1, B=2, C=3 ... Z=26

Then solve it like a number problem.

3. Word Analogy

Word analogy is different.

Here you don't normally calculate numbers.

You need to identify the **relationship between the words**.

Example:

Doctor : Hospital :: Teacher : ?

Ask:

What is the relationship between Doctor and Hospital?

Doctor works in Hospital.

Teacher works in School.

✓ Answer = School

Common Word Relationships

You should learn these categories.

1. Profession → Workplace

Doctor : Hospital :: Teacher : School

2. Person → Tool

Carpenter : Hammer :: Writer : Pen

3. Object → Function

Knife : Cutting :: Pen : Writing

4. Animal → Sound

Dog : Bark :: Cat : Meow

5. Animal → Home

Lion : Den :: Bird : Nest

6. Object → Material

Chair : Wood :: Ring : Gold

7. Cause → Effect

Exercise : Fitness :: Study : Knowledge

8. Part → Whole

Wheel : Car :: Page : Book

9. Whole → Part

Tree : Branch :: Body : Hand

10. Synonyms

Happy : Joyful :: Sad : Unhappy

11. Antonyms

Hot : Cold :: Day : Night

12. Degree/Intensity

Warm : Hot :: Cool : Cold

How to approach Word Analogy

Don't immediately look at the options.

Ask yourself:

"What exact sentence can I make using the first pair?"

For example:

Book : Author

You can say:

"An author writes a book."

Now:

Painting : ?

"An artist creates a painting."

Therefore:

Ans: **Artist**

This technique is extremely useful for difficult word analogies.

4. Mixed Analogy

Mixed analogy combines different types.

It might contain:

- Numbers + numbers
- Letters + numbers
- Letters + words

- Words + numbers
- Symbols
- Multiple relationships

These are generally more difficult because you first need to identify what type of relationship is being used.

Example 1 — Number + Number

4 : 20 :: 6 : ?

Relationship:

$$4 \times 5 = 20$$

Therefore:

$$6 \times 7 = 42$$

 **42**

Example 2 — Letter + Number

Suppose:

A : 1 :: D : ?

Alphabet position:

$$A = 1$$

$$D = 4$$

 **4**

Example 3 — Word + Word

Bird : Fly :: Fish : ?

Relationship:

Bird $\rightarrow$ Fly

Fish $\rightarrow$ Swim

 **Swim**

A Very Important Strategy

When you see:

A : B :: C : ?

Don't immediately try random calculations.

Use this **4-step process**.

Step 1 — Identify the type

Ask:

Is this a number, alphabet, word, or mixed analogy?

Step 2 — Find the relationship

For numbers:

What mathematical operation is happening?

For alphabets:

How many positions are the letters moving?

For words:

What relationship do the two words have?

Step 3 — Verify

This is extremely important.

Don't accept the first pattern you find.

Check:

Does the same relationship work exactly?

For example:

$5 \rightarrow 30$

You might think:

$\times 6$

But if the second pair doesn't follow $\times 6$, your pattern is wrong.

Step 4 — Apply to the second pair

Once the relationship is confirmed:

A → B

therefore

C → ?

Quick Cheat Sheet

Type	First thing to check
Number	+, −, ×, ÷, square, cube, combinations
Alphabet	Letter positions and forward/backward movement
Word	Meaning/relationship between the words
Mixed	Identify the type first, then apply its pattern